

Multi-Agent Reinforcement Learning

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1 Overview: Game-Theoretic Models in MARL

Multi-agent reinforcement learning draws on a rich hierarchy of game-theoretic models, each adding more structure than the last. The table below summarises the four main frameworks covered in this course.

Model	States	Agents	Key feature
Normal-Form Game	No	Finite	One-shot simultaneous play; payoff matrix
Extensive-Form Game	No	Finite	Sequential / tree-structured play; information sets
Markov Game	Yes	Finite	MDP extended to multiple agents
Mean-Field Game	Yes	$N \rightarrow \infty$	Tractable population limit

Scope of This Lecture

This lecture covers **Normal Form Games** and **Markov Games** in depth, focusing on zero-sum two-player settings.

Extensive Form Games and **Mean-Field Games** are introduced here for completeness but will *not* be studied in detail in this lecture.

2 Game Types: Brief Overview

2.1 Normal Form Games

The simplest multi-agent model: a single simultaneous interaction fully described by a payoff matrix. No time, no states, no sequential structure. Covered in depth in Section 4.

2.2 Extensive Form Games (Not Covered in This Lecture)

Extensive form games model *sequential* decision-making as a game tree.

Definition: Extensive Form Game (informal)

An extensive form game specifies:

- A **game tree** of decision nodes, chance nodes, and terminal nodes.
- At each decision node, the acting player and available actions.
- **Information sets**: what each player can observe (perfect vs. imperfect information).
- Payoffs at terminal nodes.

Key solution concept: **subgame perfect equilibrium** (backward induction).

Examples

Chess (perfect information), Poker (imperfect information), negotiation protocols.

2.3 Markov Games

The natural multi-agent extension of MDPs. Agents interact repeatedly across states. Covered in depth in Section 8.

2.4 Mean-Field Games (Not Covered in This Lecture)

Mean-field games (MFGs) study the limit of N -player symmetric games as $N \rightarrow \infty$.

Definition: Mean-Field Game (informal)

Each individual agent interacts not with specific opponents but with the **aggregate distribution** (mean field) of the entire population.

The **Mean-Field Nash Equilibrium** is a fixed point between:

1. The optimal policy of a representative agent given the population distribution, and
2. The population distribution induced by all agents following that policy.

Why Mean-Field Games? With N agents, the joint action space is exponential in N . MFGs replace this intractable interaction with a single distribution. Applications: traffic routing, financial markets, swarm robotics.

3 Unique Challenges in Multi-Agent RL

Before diving into the formal theory, it is important to understand why MARL is fundamentally harder than single-agent RL.

3.1 Non-Stationarity

In single-agent RL, the environment is stationary. In MARL, each agent's *effective environment* includes the other agents, whose policies are *also changing during learning*. This breaks standard convergence guarantees.

Challenge 1: Non-Stationarity

From Agent A's perspective, Agent B is part of the environment. But Agent B is simultaneously learning and updating its policy $\Rightarrow$ the environment is **non-stationary** from every agent's point of view.

3.2 Equilibrium Selection

Even when a Nash Equilibrium is known to exist, there may be **multiple equilibria** with different payoffs. Agents may converge to different equilibria depending on initialisation and learning dynamics.

Challenge 2: Equilibrium Selection

There is no general theory guaranteeing convergence to a *specific* equilibrium, even in cooperative settings.

3.3 Scalability of Joint Action Space

With N agents each having $|\mathcal{A}|$ actions, the joint action space has size $|\mathcal{A}|^N$.

Challenge 3: Exponential Joint Action Space

$N = 10$ agents with $|\mathcal{A}| = 5$ actions each $\Rightarrow 5^{10} \approx 10^7$ joint actions. Naïve extensions of Q-learning become intractable. Common remedies: mean-field games, factored representations, communication protocols.

3.4 Credit Assignment in Cooperative Settings

In cooperative MARL, agents share a team reward. Attributing which agent's action caused the reward, the **credit assignment problem**, is significantly harder with multiple actors.

3.5 Defining the Right Objective

In single-agent RL, the goal is unambiguous: maximise expected return. In MARL the correct objective depends on the game structure:

Setting	Objective	Notes
Zero-sum	Minimax / Nash equilibrium	Unique in value
Cooperative	Maximise joint return	Credit assignment is key
General-sum	Nash, correlated equilibrium, coarse correlated equilibrium etc.	May be non-unique

3.6 Exploration under Opponent Uncertainty

Each agent must explore to learn, but aggressive exploration can cause opponents to change behaviour, further destabilising learning.

4 Normal Form Games (Zero-Sum, In Depth)

4.1 Definition

A **normal form game** is the simplest multi-agent setting: there are *no states* and *no transitions*. It is fully specified by a utility matrix.

Definition: Normal Form Game

A two-player zero-sum normal form game is given by a utility matrix

$$R \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{B}|},$$

where

- $\mathcal{A}$ is the action set of the **max player**,
- $\mathcal{B}$ is the action set of the **min player**,
- $R(a, b)$ is the reward received by the max player (and $-R(a, b)$ by the min player).

The game is *zero-sum*: one player's gain is the other's loss.

5 Optimality Conditions in Multi-Agent RL

5.1 Why Single-Agent Optimality Fails

In single-agent RL, optimality means $V^{\pi^*}(s) \geq V^\pi(s)$ for all s and all π . This is well-defined because the environment is fixed.

In MARL, in many games, no single policy is optimal against *all* opponent strategies. Optimality is therefore *relative to the opponent*.

5.2 Optimality via Equilibrium

The standard notion of optimality in MARL is the **Nash Equilibrium**: a fixed point where no agent can gain by unilaterally deviating. It is the multi-agent counterpart of the Bellman optimality fixed point.

5.3 Nash Equilibrium

The key difference from single-agent RL: the opponent is *adaptive*, not fixed. We therefore seek an **equilibrium** rather than a single optimal policy.

Definition: Nash Equilibrium (NE)

Consider an N -player game with players $i \in \{1, \dots, N\}$, action sets $\mathcal{A}^i$, and utility functions $u^i : \mathcal{A}^1 \times \dots \times \mathcal{A}^N \rightarrow \mathbb{R}$. Let $\boldsymbol{\mu} = (\mu^1, \dots, \mu^N)$ denote a profile of mixed strategies $\mu^i \in \Delta(\mathcal{A}^i)$, and write $\boldsymbol{\mu}^{-i}$ for the strategies of all players except i .

A strategy profile $\boldsymbol{\mu}^* = (\mu^{1,*}, \dots, \mu^{N,*})$ is a **Nash Equilibrium** if for every player i and every alternative strategy $\mu^i \in \Delta(\mathcal{A}^i)$:

$$\mathbb{E}_{\mathbf{a} \sim \boldsymbol{\mu}^*} [u^i(\mathbf{a})] \geq \mathbb{E}_{\mathbf{a} \sim (\mu^i, \boldsymbol{\mu}^{-i,*})} [u^i(\mathbf{a})].$$

Informal: No player can gain by *unilaterally* deviating from their equilibrium strategy.

Two-player zero-sum special case: With $u^1 = -u^2 = R$, a NE (μ^*, ν^*) satisfies:

$$F(\mu^*, \nu) \geq F(\mu^*, \nu^*) \geq F(\mu, \nu^*) \quad \forall \mu \in \Delta(\mathcal{A}), \nu \in \Delta(\mathcal{B}).$$

Remark 1. Nash's theorem (1951) guarantees existence of a mixed-strategy NE in every finite N -player game. In two-player zero-sum games, all NE achieve the same value (the minimax value),

so the equilibrium is *unique in value* though not necessarily in strategy.

5.4 Correlated Equilibrium

The correlated equilibrium (CE) is a generalisation of the Nash Equilibrium in which players can coordinate their strategies through a shared *correlation device* (e.g. a mediator or a public signal), without requiring private independent randomisation.

Definition: Correlated Equilibrium (CE)

A **correlated equilibrium** is a joint distribution $\sigma \in \Delta(\mathcal{A}^1 \times \dots \times \mathcal{A}^N)$ over the joint action space such that, for every player i and every *deviation function* $d^i : \mathcal{A}^i \rightarrow \mathcal{A}^i$:

$$\mathbb{E}_{(a^1, \dots, a^N) \sim \sigma} [u^i(a^i, \mathbf{a}^{-i})] \geq \mathbb{E}_{(a^1, \dots, a^N) \sim \sigma} [u^i(d^i(a^i), \mathbf{a}^{-i})].$$

Mediator interpretation: A mediator samples a joint action profile $\mathbf{a} \sim \sigma$ and privately recommends a^i to each player i . The CE condition says: knowing only one's own recommendation a^i (not others'), no player benefits from deviating to $d^i(a^i)$.

Nash vs. Correlated Equilibrium

- Every NE is a CE (take $\sigma = \mu^{1,*} \otimes \dots \otimes \mu^{N,*}$, the product distribution). The converse is false in general.
- The set of CE is a convex polytope, while NE may be isolated points. CE are therefore often easier to compute (solvable by linear programming).
- CE can achieve higher social welfare than any NE (e.g. coordination games).

Example 1 (Battle of the Sexes). Two players independently choose Opera (O) or Football (F). Payoff matrix (player 1, player 2):

	O	F
O	(2, 1)	(0, 0)
F	(0, 0)	(1, 2)

There are two pure NE: (O, O) and (F, F) . A CE can assign probability $\frac{1}{2}$ to each, yielding expected payoffs $(1.5, 1.5)$ — higher than the mixed-strategy NE.

5.5 Coarse Correlated Equilibrium

Definition: Coarse Correlated Equilibrium (CCE)

Given utility matrices $U^1, U^2 \in \mathbb{R}^{|\mathcal{A}| \times |\mathcal{B}|}$ for a two-player general-sum game, a joint distribution $\pi \in \Delta(\mathcal{A} \times \mathcal{B})$ is a **Coarse Correlated Equilibrium** if:

For the max player:

$$\mathbb{E}_{(a,b) \sim \pi} [U^1(a, b)] \geq \max_{a^* \in \mathcal{A}} \mathbb{E}_{(a,b) \sim \pi} [U^1(a^*, b)] .$$

For the min player:

$$\mathbb{E}_{(a,b) \sim \pi} [U^2(a, b)] \leq \min_{b^* \in \mathcal{B}} \mathbb{E}_{(a,b) \sim \pi} [U^2(a, b^*)] .$$

Interpretation: Neither player benefits from deviating to a *fixed* action, given that the opponent continues to follow the joint distribution π . The deviation is “coarse” because the deviating player cannot condition on the sampled action recommendation.

CCE vs. Correlated Equilibrium (CE) vs. Nash

- **Nash** $\subseteq$ **CE** $\subseteq$ **CCE**: every Nash is a CE and every CE is a CCE.
- CCE allows players to commit to a joint distribution (correlated play), but deviations are to *unconditional* fixed actions (coarser than CE where deviations are conditional).
- CCE is characterised by **linear constraints** $\Rightarrow$ solvable by linear programming in polynomial time.

5.6 Social Optimum

The social optimum is *not* an equilibrium concept but rather a **centralised efficiency benchmark**: the joint strategy that maximises aggregate welfare.

Definition: Social Optimum (SO)

A strategy profile μ^{SO} is a **Social Optimum** if it maximises the *social welfare* — the sum (or weighted sum) of all players’ expected utilities:

$$\mu^{\text{SO}} = \arg \max_{\mu \in \Delta(\mathcal{A}^1) \times \dots \times \Delta(\mathcal{A}^N)} \sum_{i=1}^N \mathbb{E}_{\mathbf{a} \sim \mu} [u^i(\mathbf{a})] .$$

The corresponding **social welfare** value is:

$$\text{SW}^* = \sum_{i=1}^N \mathbb{E}_{\mathbf{a} \sim \mu^{\text{SO}}} [u^i(\mathbf{a})] .$$

Key Relationships

- The SO is generally **not** a Nash Equilibrium: selfish agents deviate from socially optimal strategies (the “tragedy of the commons”).
- The gap between the NE welfare and the SO welfare is quantified by the **Price of Anarchy** (PoA):

$$\text{PoA} = \frac{\text{SW}^*}{\min_{\mu^* \in \text{NE}} \text{SW}(\mu^*)}.$$

- In **zero-sum** games, $\sum_i u^i(\mathbf{a}) = 0$ for all $\mathbf{a}$, so the SO is trivially 0 and the concept is less useful; it is most relevant for **cooperative** and **general-sum** settings.
- A CE can achieve welfare between that of the NE and the SO, making it an attractive middle ground.

5.7 Comparison of Solution Concepts

Concept	Incentive condition	Coordination	Welfare
Nash Equilibrium	No unilateral deviation	None (independent)	Can be low (PoA)
Correlated Equil.	No unilateral deviation given recommendation	Via mediator	$\geq$ NE welfare
Social Optimum	No incentive condition	Centralised planner	Maximum

5.8 Pure vs. Mixed Strategies

Definition: Pure Strategy

A **pure strategy** commits a player to a single deterministic action.

Key Observation

A pure-strategy NE may *not* always exist.

Example — Rock Paper Scissors:

$$R = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix} \quad (\text{rows: Rock, Paper, Scissors})$$

Computing the minimax values:

$$\max_a \min_b R(a, b) = -1, \quad \min_b \max_a R(a, b) = 1.$$

Since $-1 \neq 1$, the minimax equality fails $\Rightarrow$ **no pure-strategy NE exists.**

Definition: Mixed Strategy

A **mixed strategy** is a probability distribution over actions:

$$\mu \in \Delta(\mathcal{A}), \quad \nu \in \Delta(\mathcal{B}).$$

The expected utility under mixed strategies is the *bilinear* function

$$F(\mu, \nu) = \sum_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} \mu(a) R(a, b) \nu(b) = \mu^\top R \nu.$$

6 Minimax Theorem

Theorem: von Neumann Minimax Theorem

Let $\mathcal{U}, \mathcal{V}$ be convex compact sets and $F : \mathcal{U} \times \mathcal{V} \rightarrow \mathbb{R}$ be *concave-convex* (concave in the first argument, convex in the second). Then

$$\max_{\mu \in \mathcal{U}} \min_{\nu \in \mathcal{V}} F(\mu, \nu) = \min_{\nu \in \mathcal{V}} \max_{\mu \in \mathcal{U}} F(\mu, \nu).$$

Furthermore, there exists a saddle-point (μ^*, ν^*) achieving both optima.

Remark 2. $F(\mu, \nu) = \mu^\top R \nu$ is bilinear, hence concave-convex. The strategy simplices $\Delta(\mathcal{A}), \Delta(\mathcal{B})$ are convex and compact. Therefore the minimax theorem applies to every normal form game.

Corollary 1. A *mixed-strategy Nash Equilibrium always exists* in every finite two-player zero-sum normal form game.

Example 2 (Rock Paper Scissors — Mixed NE). The unique NE is the *uniform* distribution:

$$\mu^* = \nu^* = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right).$$

Under this strategy, regardless of the opponent's play, every player achieves expected utility 0.

7 Best Response

Definition: Best Response

The **best response** is the optimal strategy *given* a fixed opponent strategy.

For the max player against fixed ν :

$$\text{BR}_{\max}(\nu) = \arg \max_{\mu} F(\mu, \nu).$$

For the min player against fixed μ :

$$\text{BR}_{\min}(\mu) = \arg \min_{\nu} F(\mu, \nu).$$

7.1 Nash Equilibrium via Best Response

Proposition: Second Characterisation of Nash

(μ^*, ν^*) is a Nash Equilibrium if and only if each player plays a best response to the other:

$$\mu^* = \text{BR}_{\max}(\nu^*), \quad \nu^* = \text{BR}_{\min}(\mu^*).$$

Equivalently, the NE solves

$$F(\mu^*, \nu^*) = \max_{\mu} F(\mu, \text{BR}_{\min}(\mu)) = \min_{\nu} F(\text{BR}_{\max}(\nu), \nu).$$

Interpretation: If your opponent *always* plays the best response to you, the NE is your optimal strategy.

8 Markov Games (Zero-Sum, In Depth)

8.1 Definition

A **Markov game** (or stochastic game) extends the normal form game to settings with states and transitions — i.e. it is the multi-agent analogue of an MDP.

Definition: Two-Player Zero-Sum Markov Game

A Markov game is a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{B}, P, r, H)$:

- $\mathcal{S}$: state space
- $\mathcal{A}$: action space of the max player
- $\mathcal{B}$: action space of the min player
- $P : \mathcal{S} \times \mathcal{A} \times \mathcal{B} \rightarrow \Delta(\mathcal{S})$: transition kernel
- $r : \mathcal{S} \times \mathcal{A} \times \mathcal{B} \rightarrow \mathbb{R}$: reward function
- H : episode horizon

Both P and r depend on *both* players' actions, unlike the single-agent MDP.

8.2 Policies and Values

Policies are *randomised* (necessary to guarantee NE existence, as in the normal form case):

$$\mu = \{\mu_h : \mathcal{S} \rightarrow \Delta(\mathcal{A})\}_{h=1}^H, \quad \nu = \{\nu_h : \mathcal{S} \rightarrow \Delta(\mathcal{B})\}_{h=1}^H.$$

Remark 3. Policies condition only on the *current state*, not the opponent's action. This models **simultaneous-move** gameplay: both players act at the same time.

The value function for a fixed policy pair (μ, ν) :

$$V_h^{\mu, \nu}(s) = \mathbb{E} \left[\sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'}, b_{h'}) \mid s_h = s, a \sim \mu, b \sim \nu \right].$$

The Q -function:

$$Q_h^{\mu, \nu}(s, a, b) = r_h(s, a, b) + \mathbb{E}_{s' \sim P(\cdot | s, a, b)} [V_{h+1}^{\mu, \nu}(s')].$$

8.3 Bellman Equations for Fixed Policy

$$Q_h^{\mu,\nu}(s, a, b) = r_h(s, a, b) + \sum_{s'} P(s'|s, a, b) V_{h+1}^{\mu,\nu}(s'),$$

$$V_h^{\mu,\nu}(s) = \sum_a \sum_b \mu_h(a|s) \nu_h(b|s) Q_h^{\mu,\nu}(s, a, b).$$

9 Optimal Strategies in Markov Games

9.1 Best Response in Markov Games

Theorem: Best Response in Markov Games

For any fixed policy μ of the max player, there exists a best-response policy $\nu^\dagger(\mu)$ of the min player such that for *all* states s and steps h *simultaneously*:

$$V_h^{\mu, \nu^\dagger(\mu)}(s) = \min_{\nu} V_h^{\mu, \nu}(s).$$

Why? When μ is fixed, the Markov game reduces to an MDP from the min player's perspective. Standard MDP theory then guarantees the existence of a simultaneously optimal policy.

We use the shorthand:

$$V_h^{\mu, \dagger}(s) := V_h^{\mu, \nu^\dagger(\mu)}(s).$$

9.2 Nash Equilibrium in Markov Games

Theorem: Nash Equilibrium Existence (Markov Games)

For any two-player zero-sum Markov game, there exists a Nash Equilibrium (μ^*, ν^*) such that for all s, h :

$$V_h^{\mu^*, \nu^*}(s) = \max_{\mu} V_h^{\mu, \nu^\dagger(\mu)}(s) = \min_{\nu} V_h^{\mu^\dagger(\nu), \nu}(s).$$

Furthermore, the minimax equality holds:

$$\max_{\mu} \min_{\nu} V_h^{\mu, \nu}(s) = \min_{\nu} \max_{\mu} V_h^{\mu, \nu}(s).$$

9.3 Optimal Bellman Equations

$$Q_h^*(s, a, b) = r_h(s, a, b) + \sum_{s'} P(s'|s, a, b) V_{h+1}^*(s'),$$

$$V_h^*(s) = \underbrace{\min_{\nu} \max_{\mu}}_{\text{or } \max_{\mu} \min_{\nu}} \sum_{a,b} \mu(a) \nu(b) Q_h^*(s, a, b).$$

Remark 4. The only difference from the single-agent Bellman equation is that $\max_a$ is replaced by $\min_{\nu} \max_{\mu}$ (a matrix game solve). This reduces to a linear program at each state.

10 Measuring Suboptimality: Duality Gap

Since the true NE is never found exactly in practice, we need a metric to measure how close a policy pair (μ, ν) is to equilibrium.

Definition: Duality Gap

The **duality gap** of a policy pair (μ, ν) at state s_1 is:

$$\text{DualityGap}(\mu, \nu) = V_1^{\mu^\dagger(\nu), \nu}(s_1) - V_1^{\mu, \nu^\dagger(\mu)}(s_1).$$

Intuitively: how much can the min player exploit μ , minus how much can the max player exploit ν .

Proposition

(μ, ν) is a Nash Equilibrium if and only if $\text{DualityGap}(\mu, \nu) = 0$.

We say a policy pair (μ, ν) is ε -optimal (an ε -**Nash Equilibrium**) if:

$$\text{DualityGap}(\mu, \nu) \leq \varepsilon.$$

11 Planning Algorithms

Planning assumes the transition P and reward r are **known**.

11.1 Finding the Best Response (Planning)

When one player is fixed, this reduces to an MDP. Standard value iteration applies directly.

11.2 Finding the Nash Equilibrium (Planning)

Run **Nash Value Iteration**: backward induction from $h = H$ to $h = 1$.

Algorithm: Nash Value Iteration

For $h = H, H - 1, \dots, 1$:

1. Update Q -values:

$$Q_h^*(s, a, b) = r_h(s, a, b) + \sum_{s'} P(s'|s, a, b) V_{h+1}^*(s').$$

2. Solve the matrix game at each state s :

$$(\mu_h^*(s), \nu_h^*(s)) = \text{NashSolve}(Q_h^*(s, \cdot, \cdot)).$$

3. Update V -values using the Nash strategies:

$$V_h^*(s) = \sum_{a,b} \mu_h^*(s, a) \nu_h^*(s, b) Q_h^*(s, a, b).$$

Remark 5. Step 2 at each state is a *normal form game* solve, which can be done via linear programming in polynomial time.

Concept	Normal Form Game	Markov Game
State space	None	$\mathcal{S}$
Transition	None	$P(s' s, a, b)$
Pure NE	May not exist	May not exist
Mixed NE	Always exists (von Neumann)	Always exists (DP argument)
Optimality criterion	Minimax value	Minimax value function
Algorithm	LP solve	Nash Value Iteration

12 Best-Response Based Algorithms

Previously, we established the formulation of Markov games and showed how to *plan* using the Bellman optimality equation (Nash-VI) when the model is known. Now we turn to **learning algorithms**, i.e. what to do when the transition and reward are unknown and must be estimated from data.

12.1 Self-Play

Algorithm: Self-Play

Initialise policies π_1^1, π_2^1 arbitrarily. For $k = 1, 2, \dots, K$:

$$\pi_1^{k+1} = \text{BR}(\pi_2^k), \quad \pi_2^{k+1} = \text{BR}(\pi_1^{k+1}).$$

Each player computes the best response to the *current* policy of the opponent.

Why Self-Play Fails to Converge

The fundamental problem is that self-play is a *greedy* procedure with no memory. At each iteration, the current policy is discarded entirely in favour of the best response to the latest opponent. This creates an adversarial feedback loop.

Problem: Self-Play Does Not Converge

On Rock-Paper-Scissors, self-play cycles deterministically:

$$\text{Rock} \xrightarrow{\text{BR}} \text{Paper} \xrightarrow{\text{BR}} \text{Scissors} \xrightarrow{\text{BR}} \text{Rock} \xrightarrow{\text{BR}} \dots$$

The iterates cycle with period 3 and never approach the Nash equilibrium $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

More generally, in any game where the best-response correspondence is not a contraction, self-play can exhibit periodic orbits, quasi-periodic drift, or chaotic behaviour.

When Does Self-Play Work?

Despite the failure on zero-sum games in general, self-play does converge in specific structured settings.

Theorem: Self-Play Convergence in Potential Games

In a **potential game**, i.e. a game admitting a function $\Phi : \mathcal{A}^1 \times \dots \times \mathcal{A}^N \rightarrow \mathbb{R}$ such that for all i ,

$$u^i(a^{i'}; \mathbf{a}^{-i}) - u^i(a^i; \mathbf{a}^{-i}) = \Phi(a^{i'}; \mathbf{a}^{-i}) - \Phi(a^i; \mathbf{a}^{-i}),$$

best-response self-play converges to a pure Nash Equilibrium in finite time, since each update strictly increases Φ and Φ is bounded.

Self-Play in Practice: AlphaGo / AlphaZero Despite the theoretical failure case, **approximate** self-play powered by deep neural networks has achieved superhuman performance in Go [Silver et al., 2017] and Chess [Silver et al., 2018]. The key differences from the theoretical setting:

- Policies are parameterised by neural networks with limited capacity, preventing exact best responses and breaking the cycling argument.
- The algorithm uses **MCTS** (Monte Carlo Tree Search) for approximate best-response computation.
- A **replay buffer** retains historical self-play games, providing implicit averaging similar to fictitious play.

The theoretical explanation for why approximate self-play succeeds in large games remains an active research area.

12.2 Fictitious Play

Proposed by [Brown, 1951]. The key fix: compute best response to the *average* opponent policy, not the current one.

Algorithm: Fictitious Play

For $k = 1, 2, \dots, K$:

$$\pi_1^{k+1} = \text{BR} \left(\frac{1}{k} \sum_{j=1}^k \pi_2^j \right), \quad \pi_2^{k+1} = \text{BR} \left(\frac{1}{k+1} \sum_{j=1}^{k+1} \pi_1^j \right).$$

Each player best-responds to the **empirical average** of the opponent's past policies.

Convergence

Theorem: Convergence of Fictitious Play [Robinson, 1951]

If the players of a zero-sum game $(R, C = -R)$ interact via fictitious play, then

$$\lim_{t \rightarrow \infty} \max_i e_i^\top R y_t = \lim_{t \rightarrow \infty} \min_j x_t^\top R e_j = v,$$

where v is the value of the row player. In particular, both limits exist.

Corollary 2 (Empirical Strategies Converge to Nash). *For any $\varepsilon > 0$, there exists T such that for all $t \geq T$, the empirical strategy pair (x_t, y_t) is an ε -approximate Nash Equilibrium.*

The convergence requires $\mathcal{O}\left(\frac{1}{\varepsilon}\right)^{(A)}$ to converge.

12.2.1 Example on RPS

Rock-Paper-Scissors is an ideal test case: it has a unique NE (the uniform distribution), no pure-strategy NE, and is the canonical game where naive self-play cycles. Fictitious play must therefore eventually “mix” its empirical strategy through accumulated history rather than any single best response.

The payoff matrix for the row player is

$$R = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix} \quad (\text{rows/columns: Rock, Paper, Scissors}).$$

The game value is $v = 0$ and the unique NE is $\mu^* = \nu^* = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.

First few rounds. Suppose $i_1 = R$ (Rock), $j_1 = R$ (Rock). The cumulative payoff vectors and selected actions evolve as follows.

t	i_t	j_t	$t e_R^\top R y_t$	$t e_P^\top R y_t$	$t e_S^\top R y_t$	$t x_t^\top C e_R$	$t x_t^\top C e_P$	$t x_t^\top C e_S$
1	R	R	<u>0</u>	1	-1	<u>0</u>	-1	1
2	P	S	-1	<u>2</u>	0	1	-1	<u>0</u>
3	P	S	-2	<u>3</u>	1	2	-2	<u>-1</u>
4	P	S	-3	<u>4</u>	2	3	-3	<u>-2</u>
5	P	R	-4	<u>4</u>	3	<u>3</u>	-3	-3
6	S	R	-4	4	<u>4</u>	<u>4</u>	-3	-4

What to observe.

- Unlike self-play, the players do *not* simply chase the opponent’s last action. Because they best-respond to the *accumulation* of past play, the pressure to switch action builds slowly.
- The empirical strategies x_t and y_t do not converge monotonically: the column player plays Scissors three times in a row (rounds 2–4), which inflates the cumulative payoff for Paper, before eventually switching to Rock at round 5.
- **Self-play:** iterates cycle $R \rightarrow P \rightarrow S \rightarrow R \rightarrow \dots$ and *never* converge.

- **Fictitious play:** iterates also cycle in pure-strategy selections (as seen above), but the *empirical averages* x_t, y_t converge to the Nash equilibrium $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$.
- The key is that history accumulates: once a strategy has been over-played, it takes many future rounds to shift the empirical average, naturally damping the cycles.

12.3 Double Oracle

Double Oracle [McMahan et al., 2003] is a smarter variant that maintains a *restricted* strategy set and iteratively expands it.

Algorithm: Double Oracle

Maintain strategy sets $\Pi_1^k \subseteq \Pi_1$ and $\Pi_2^k \subseteq \Pi_2$. At iteration k :

1. Compute the Nash equilibrium (p, q) of the restricted **meta-game** matrix M^k given by Π_1^k and Π_2^k .
2. Expand:

$$\pi_1^{k+1} = \text{BR} \left(\sum_i^k q_i \pi_2^i \right), \quad \pi_2^{k+1} = \text{BR} \left(\sum_i^k p_i \pi_1^i \right).$$

3. Add π_1^{k+1} and π_2^{k+1} to the restricted sets; update M^k .

Proposition: Double Oracle Correctness and Complexity

Double Oracle terminates in at most $|\mathcal{A}|$ iterations and outputs the exact Nash Equilibrium of the normal-form game.

Why? At each step, the BR oracle adds a strategy that is a pure best response to the current meta-NE. Once no player has a profitable pure-strategy deviation, the restricted NE is also a NE of the full game (by the support lemma: at a NE, all actions in the support must be best responses).

Remark 6. The $\mathcal{O}(|\mathcal{A}|)$ bound is tight in the worst case (all actions may be needed), but in practice Double Oracle often terminates in far fewer iterations because many actions are never best responses to any mixture.

12.4 Policy-Space Response Oracles (PSRO)

PSRO [Lanctot et al., 2017] generalises Double Oracle from normal-form games to Markov games with function approximation. It replaces the exact best-response and exact NE-solve steps with approximate counterparts, making the framework practical for large games.

The General PSRO Framework

Algorithm: PSRO

- Maintain a **population** (strategy set) $\Pi^k = \{\pi^1, \dots, \pi^k\}$ for each player. At iteration k :
1. **Meta-game evaluation.** For all strategy pairs $(\pi_1^i, \pi_2^j) \in \Pi_1^k \times \Pi_2^k$, estimate the payoff $M_{ij}^k \approx F(\pi_1^i, \pi_2^j)$ by simulation.
 2. **Meta-strategy solve.** Compute a *meta-strategy* $(p^k, q^k) \in \Delta(\Pi_1^k) \times \Delta(\Pi_2^k)$ using a **meta-solver** $\mathcal{M}$ applied to M^k . Common choices:
 - Nash (exact, but computationally expensive for large populations),
 - Uniform mixture,
 - Projected Replicator Dynamics (PRD),
 - α -Rank (ranking-based; does not require NE computation).
 3. **Oracle call.** Compute approximate best responses against the current meta-strategies:

$$\pi_1^{k+1} \approx \text{BR} \left(\sum_j q_j^k \pi_2^j \right), \quad \pi_2^{k+1} \approx \text{BR} \left(\sum_i p_i^k \pi_1^i \right).$$

In Markov games, this is a single-agent RL problem (e.g. solved by PPO, DQN).

4. **Expand population.** Add π_1^{k+1} and π_2^{k+1} to the respective populations.

Output: the meta-strategy (p^K, q^K) as the final (mixed) policy.

Why PSRO Generalises Double Oracle

Double Oracle is the special case of PSRO where:

- The meta-solver $\mathcal{M}$ computes the exact Nash of M^k , and
- The oracle computes exact best responses.

PSRO replaces both with approximate counterparts, enabling application to games where exact computation is infeasible.

The Role of the Meta-Solver

The choice of meta-solver $\mathcal{M}$ fundamentally shapes the behaviour and convergence of PSRO.

Meta-solver	Convergence guarantee	Notes
Nash	Exact NE (if oracle is exact)	PPAD-hard for general-sum; LP for zero-sum
Uniform	No NE guarantee	Simple; diverse exploration
Replicator dynamics	Converges to NE in zero-sum	Slow; sensitive to initialisation
α -Rank	Stationary distribution of random walk	No NE guarantee; robust; no LP needed

α -Rank [Omidshafiei et al., 2019] defines a Markov chain over strategy profiles where transitions favour strategies that are better responses. Its stationary distribution ranks strategies by evolutionary dominance. It avoids NE computation entirely and is particularly suited to *many-player* and *general-sum* games where Nash is intractable.

AlphaStar [Vinyals et al., 2019]

AlphaStar, the first agent to achieve Grandmaster level at StarCraft II, uses a PSRO-style league training framework:

- A **population** of agents (main agents, league exploiters, main exploiters) is maintained.
- Each agent is trained via RL (V-trace + self-imitation) against a mixture of opponents drawn from the population according to a PFSP (Prioritised Fictitious Self-Play) meta-strategy.
- The mixture prioritises opponents against which the agent has win rate closest to 50%, providing a curriculum of well-matched adversaries.
- The population is periodically expanded by adding the current policy as a new fixed “historical” agent.

This directly implements the PSRO loop with a learned RL oracle and a win-rate-adaptive meta-solver.

Limitations

The two fundamental issues identified for BR-based methods in Markov games persist in PSRO, and are worth restating precisely:

Issue 1: Mixture of Markov Policies is Non-Markovian The meta-strategy $\sum_i p_i \pi^i$ is a *mixture over policies*, not a single policy. A mixture of Markov policies requires tracking *which* policy is being executed (the “mixture index”), making it a history-dependent strategy of exponential complexity. **Consequence:** The single-agent RL oracle is asked to compute a best response against a non-Markovian opponent, which standard RL algorithms (designed for MDPs with fixed transition kernels) are not guaranteed to solve correctly. The environment seen by the oracle is non-stationary.

Issue 2: Exponential Policy Space The strategy space of a Markov game (policies mapping states to action distributions) is exponentially large in the state space. The meta-game matrix M^k has entries indexed by these policies, so even $K = \mathcal{O}(|\Pi|)$ iterations are intractable. In contrast, Nash-VI and Optimistic Nash-VI exploit the Markov structure directly and achieve polynomial sample complexity.

13 Dynamic Programming Based Approach

13.1 Generative Model Setting

In the **generative model** [Liu et al., 2021] (simulator) setting, one can query any (s, a, b) to obtain a sample of the next state and reward.

Nash Value Iteration with Empirical Model

1. **Collect samples:** Sample (s, a, b, s', r) uniformly to estimate the empirical transition $\hat{P}_h(s' | s, a, b)$.
2. **Run Nash-VI** with $\hat{P}$ in place of P :

Algorithm: Nash-VI (Generative Model)

For $h = H, H - 1, \dots, 1$, for all (s, a, b) :

$$\hat{Q}_h(s, a, b) = r_h(s, a, b) + \hat{P}_h V_{h+1}(s, a, b),$$

$$V_h(s) = \min_{\nu} \max_{\mu} \sum_{a,b} \mu(a) \nu(b) \hat{Q}_h(s, a, b).$$

The inner optimisation is a normal-form game solve (LP).

13.2 Exploration Setting: Optimistic Nash Value Iteration

In the online exploration setting, the agent must collect its own data by interacting with the game. The key principle, as in single-agent RL, is optimism in the face of uncertainty.

Because there are two players with opposing objectives, optimism must be applied in *both directions*:

- **Upper bound $\bar{Q}_h$:** serves as optimism for the *max* player.
- **Lower bound $\underline{Q}_h$:** serves as optimism for the *min* player.

Algorithm: Optimistic Nash-VI

For episode $k = 1, \dots, K$; for $h = H, \dots, 1$; for all (s, a, b) :

Step 1 — Compute optimistic Q-values:

$$\bar{Q}_h^k(s, a, b) = r_h(s, a, b) + \hat{P}_h^k \bar{V}_{h+1}^k(s, a, b) + \beta_h^k(s, a, b),$$

$$\underline{Q}_h^k(s, a, b) = r_h(s, a, b) + \hat{P}_h^k \underline{V}_{h+1}^k(s, a, b) - \beta_h^k(s, a, b).$$

Step 2 — Compute V-values via CCE (see Section ??):

$$\pi_h^k(\cdot | s) = \text{CCE}\left(\bar{Q}_h^k(s, \cdot, \cdot), \underline{Q}_h^k(s, \cdot, \cdot)\right),$$

$$\bar{V}_h^k(s) = \mathbb{E}_{(a,b) \sim \pi_h^k(\cdot | s)} \left[\bar{Q}_h^k(s, a, b) \right], \quad \underline{V}_h^k(s) = \mathbb{E}_{(a,b) \sim \pi_h^k(\cdot | s)} \left[\underline{Q}_h^k(s, a, b) \right].$$

Step 3 — Execute π^k , collect one trajectory, update $\hat{P}^k$.

13.3 Why Not Nash?

A naive implementation of Optimistic Nash-VI would compute a Nash equilibrium of the *general-sum* game with utilities $(\bar{Q}_h^k, \underline{Q}_h^k)$ (since $\bar{Q} \neq -\underline{Q}$ in general). However:

Computing a Nash equilibrium of a general-sum game is **PPAD-complete** — no known polynomial-time algorithm exists in the worst case. This is a fundamental computational bottleneck, even for

the tabular setting.

13.4 Why CCE Suffices for Zero-Sum Markov Games

The critical property that justifies replacing Nash by CCE in the algorithm is the following:

Proposition: CCE Marginals are Nash in Zero-Sum Games

Let $\pi \in \Delta(\mathcal{A} \times \mathcal{B})$ be a CCE of the *zero-sum* game with utility Q (i.e. $U^1 = Q$ and $U^2 = -Q$). Define the marginals:

$$\mu(a) = \sum_b \pi(a, b), \quad \nu(b) = \sum_a \pi(a, b).$$

Then (μ, ν) is a **Nash Equilibrium** of the zero-sum game with utility Q .

In Optimistic Nash-VI, as the number of samples grows, $\bar{Q}^k \approx \underline{Q}^k \approx Q^*$. In the limit, the CCE of $(\bar{Q}^k, \underline{Q}^k)$ approaches a CCE of a zero-sum game, whose marginals are Nash. This is the key insight that makes the algorithm principled.

14 Offline Multi-Agent Reinforcement Learning [Cui and Du, 2022]

In previous lectures, we studied the **online** setting, where agents actively interact with the environment to collect data. In the **offline** setting, the agent has access only to a *fixed, pre-collected dataset* and cannot query the environment further.

Definition: Offline MARL

Given a fixed dataset

$$\mathcal{D} = \{(s_h^\tau, a_h^\tau, b_h^\tau, r_h^\tau, s_{h+1}^\tau)\}_{\tau \in [n], h \in [H]},$$

collected by some (possibly unknown) exploration policy ρ , the goal of **offline two-player zero-sum Markov game** learning is to find an ε -approximate Nash Equilibrium strategy using only $\mathcal{D}$, without any further environment interaction.

14.1 Why Offline MARL?

Offline learning is important whenever:

- Online interaction is expensive, dangerous, or unethical (e.g. healthcare, autonomous driving, robotics).
- Only historical logs of multi-agent interactions are available.
- Re-deploying updated policies mid-game is impractical.

The central challenge is **distribution shift**: the dataset was collected under ρ , but we want to evaluate and optimise policies (μ, ν) that may visit very different state-action regions.

14.2 Recap: Offline Single-Agent RL, the Pessimism Principle

In offline single-agent RL, the key algorithmic idea is **pessimism in the face of uncertainty**: when the dataset provides little information about a state–action pair, be *conservative* (pessimistic) rather than optimistic about its value. This discourages the policy from exploiting regions not covered by the data.

Known Result: Offline Single-Agent RL

Under the **single policy concentration** assumption, the dataset covers the optimal policy π^* , pessimism-based algorithms (LCB-VI and variants) find an ε -optimal policy with sample complexity

$$n = \tilde{\mathcal{O}}\left(\frac{S C^* H^3}{\varepsilon^2}\right),$$

where $C^* = \max_{h,s,a} \frac{d_h^{\pi^*}(s,a)}{d_h^{\rho}(s,a)}$ is the *single-policy concentrability coefficient* quantifying how well the optimal policy is covered by the dataset.

Remark 7. $d_h^{\pi^*}(s,a)$ is the **state-action occupancy measure** of the optimal policy π^* at step h . In a finite-horizon MDP with horizon H , starting from a fixed initial state s_1 (or initial distribution ρ_0), it is defined as

$$d_h^{\pi^*}(s,a) := \Pr(s_h = s, a_h = a \mid \pi^*, s_1),$$

and decomposes as

$$d_h^{\pi^*}(s,a) = d_h^{\pi^*}(s) \cdot \pi_h^*(a \mid s),$$

where $d_h^{\pi^*}(s) = \Pr(s_h = s \mid \pi^*, s_1)$ is the marginal state distribution at step h under π^* . These distributions satisfy the flow equations:

$$d_0^{\pi^*}(s) = \mathbf{1}[s = s_0], \quad d_{h+1}^{\pi^*}(s') = \sum_{s,a} d_h^{\pi^*}(s,a) P_h(s' \mid s,a), \quad h = 1, \dots, H-1.$$

The ratio $d_h^{\pi^*}(s,a)/d_h^{\rho}(s,a)$ measures how much π^* visits (s,a) at step h relative to how often the offline dataset covers it. Taking the maximum over all (h,s,a) gives $C^* = \max_{h,s,a} d_h^{\pi^*}(s,a)/d_h^{\rho}(s,a)$: a small C^* means the dataset covers π^* well at every step; a large C^* indicates there exist triples (h,s,a) that π^* visits often but the data barely reaches.

Key Point for Single-Agent RL

Single policy concentration, covering *only* the optimal policy π^* , is **both necessary and sufficient** for offline single-agent RL. This is the minimal possible coverage assumption.

The natural question is: does the same hold for offline MARL? Is it sufficient to cover only the Nash Equilibrium strategy (μ^*, ν^*) ?

14.3 Dataset Coverage Assumptions for Markov Games

We define two coverage assumptions in order of increasing strength.

Assumption 1: Single Strategy Concentration

The NE strategy (μ^*, ν^*) is covered by the dataset:

$$\frac{d_h^{\mu^*, \nu^*}(s, a, b)}{d_h^\rho(s, a, b)} < \infty \quad \forall h, (s, a, b) \text{ with } d_h^{\mu^*, \nu^*}(s, a, b) > 0.$$

Analogue of single policy concentration in single-agent RL.

Assumption 2: Unilateral Concentration

For *all* strategies μ and ν , both (μ^*, ν) and (μ, ν^*) are covered by the dataset:

$$\frac{d_h^{\mu^*, \nu}(s, a, b)}{d_h^\rho(s, a, b)} < \infty \quad \text{and} \quad \frac{d_h^{\mu, \nu^*}(s, a, b)}{d_h^\rho(s, a, b)} < \infty \quad \forall \mu, \nu, h, (s, a, b).$$

The dataset must cover every strategy paired with the NE opponent. The concentrability coefficient is:

$$C^* = \max \left\{ \max_{\mu, h, s, a, b} \frac{d_h^{\mu^*, \nu^*}(s, a, b)}{d_h^\rho(s, a, b)}, \max_{\nu, h, s, a, b} \frac{d_h^{\mu^*, \nu}(s, a, b)}{d_h^\rho(s, a, b)} \right\}.$$

14.4 The Hardness Result: Single Strategy Concentration is Insufficient

The main surprise of offline MARL is that the analogue of the single-agent sufficient condition *fails*:

Theorem: Hardness of Offline MARL [Cui and Du, 2022]

There exist two Markov game instances $\mathcal{G}_1$ and $\mathcal{G}_2$ such that:

1. Both instances satisfy **single strategy concentration** (the NE strategy is covered by the dataset).
2. The two instances are **statistically indistinguishable** given any dataset of polynomial size generated under the same exploration policy.
3. The NE strategies of $\mathcal{G}_1$ and $\mathcal{G}_2$ are **different**.

Therefore, no algorithm can learn the NE strategy under single strategy concentration alone, regardless of the sample size.

14.5 Intuition: Why Single Concentration Fails

In single-agent RL, we only need to evaluate the *optimal policy* against a fixed environment. The pessimism principle ensures we do not venture into uncovered regions.

In MARL, to verify that (μ^*, ν^*) is a Nash Equilibrium, we must check that *no unilateral deviation is profitable*. For the max player, this means verifying:

$$V_1^{\mu^*, \nu^*}(s_1) \geq V_1^{\mu, \nu^*}(s_1) \quad \forall \mu.$$

This requires evaluating (μ, ν^*) for *every* possible μ — pairs that may not be covered by single strategy concentration. Without coverage of these off-NE strategy pairs, we cannot distinguish between game instances that agree on the NE trajectory but differ elsewhere.

The Core Difficulty

Single-agent optimality is a *self-certifying* property: if we know π^* is covered, pessimism ensures we identify it. Nash optimality is *not* self-certifying: verifying that μ^* is optimal requires knowing what would happen under *any* deviation μ , which the NE trajectory alone does not reveal.

	Single-Agent RL	Offline MARL (Zero-Sum)
Minimal coverage	Single policy concentration	Unilateral concentration
Self-certifying?	Yes	No
Pessimism sufficient?	Yes (under single concentration)	Only under unilateral concentration
Key challenge	Distribution shift	Distribution shift <i>and</i> deviation coverage

14.6 Positive Result: Pessimism under Unilateral Concentration

Theorem: Unilateral Concentration is Necessary

For any $\varepsilon > 0$, there exists a family of Markov game instances such that if the unilateral concentration assumption is *even slightly violated* (i.e. there exists some μ and a set of states reachable by (μ, ν^*) but not covered by $\mathcal{D}$), then no algorithm can output an ε -approximate NE with probability greater than 1/2, regardless of sample size.

Combined with the positive result above, this gives a complete characterisation:

Corollary: Necessary and Sufficient Condition

Unilateral concentration is the necessary and sufficient dataset coverage assumption for offline two-player zero-sum Markov game learning.

- **Sufficient:** Pessimistic Nash-VI achieves polynomial sample complexity.
- **Necessary:** Any algorithm fails without it.

15 From Offline MARL to Imitation Learning [Freihaut et al., 2026]

15.1 What Changes?

Previously we studied offline MARL: we had a dataset of transitions (s, a, b, r, s') and the goal was to find a near-Nash equilibrium. The reward function was *known* as part of the data.

Imitation Learning (IL) drops this assumption. The learner has access only to *expert trajectories*, i.e., sequences of states and actions generated by an expert playing a Nash Equilibrium, but *no reward signal is observed*. The task is still to recover an approximate Nash Equilibrium, but now purely from behavioural data.

Definition: Multi-Agent Imitation Learning (MAIL)

Let (μ^E, ν^E) be an expert Nash Equilibrium strategy. In Multi-Agent Imitation Learning, the learner observes trajectories generated by (μ^E, ν^E) but does *not* observe reward signals. The goal is to output $(\hat{\mu}, \hat{\nu})$ such that:

$$\mathbb{E}[\text{Nash-Gap}(\hat{\mu}, \hat{\nu})] < \varepsilon,$$

where $\text{Nash-Gap}(\mu, \nu) := \langle d_0, V_0^{\mu^*, \nu} - V_0^{\mu, \nu^*} \rangle \geq 0$, with equality if and only if (μ, ν) is a Nash Equilibrium.

15.2 Two Settings

Setting 1: Non-Interactive MAIL

The learner receives a **fixed pre-collected dataset** of expert trajectories. States are sampled from some distribution $\rho_h \in \Delta(\mathcal{S})$; actions are queried from the expert at those states:

$$\{s_h^i\}_{i=1}^N \sim \rho_h, \quad a_h^i \sim \mu^E(\cdot | s_h^i), \quad b_h^i \sim \nu^E(\cdot | s_h^i).$$

No further interaction with the expert is allowed after the dataset is received.

Setting 2: Interactive MAIL

The learner can **query the expert on demand** at any state visited during training. The expert answers with $(\mu^E(\cdot | s), \nu^E(\cdot | s))$ for any queried state s . The learning process alternates between environment interaction and expert queries.

Key Distinction

Non-interactive MAIL is weaker (no queriable expert) but applicable in more real-world scenarios. Interactive MAIL is more powerful but requires access to an expert that can be queried at arbitrary states, which is a strong assumption not always available in practice.

15.3 Why MAIL Differs from Offline MARL

At first glance, MAIL looks like a special case of offline MARL: we have a dataset and no further interaction. However, there is a **fundamental difference in the coverage requirement**.

15.4 Recap: Coverage in Offline MARL

In offline MARL, the key coverage notion was **unilateral concentration**: for all μ and ν , the pairs (μ, ν^*) and (μ^*, ν) must be covered by the dataset. The concentrability coefficient was:

$$C^* = \max \left\{ \max_{\mu, h, s, a, b} \frac{d_h^{\mu, \nu^*}(s, a, b)}{d_h^\rho(s, a, b)}, \max_{\nu, h, s, a, b} \frac{d_h^{\mu^*, \nu}(s, a, b)}{d_h^\rho(s, a, b)} \right\}.$$

This measures how well the dataset covers deviations paired with the *Nash opponent*.

15.5 Coverage in MAIL: Two Coefficients

In MAIL, the dataset is generated by the expert (μ^E, ν^E) and there is no reward. Two coverage coefficients are relevant:

Definition: Expert Concentrability Coefficient

$$\mathcal{C}(\mu^E, \nu^E) := \max \left\{ \max_{\nu^* \in \text{br}(\mu^E), h} \left\| \frac{d_h^{\mu^E, \nu^*}}{\rho_h} \right\|_{\infty}, \max_{\mu^* \in \text{br}(\nu^E), h} \left\| \frac{d_h^{\mu^*, \nu^E}}{\rho_h} \right\|_{\infty} \right\}.$$

This measures coverage of states reachable by **rational deviations** — best responses to the expert policy. These are the deviations a Nash opponent would make.

Definition: All-Policy Concentrability Coefficient

$$\mathcal{C}_{\max} := \max_{\mu, \nu} \mathcal{C}(\mu, \nu).$$

This measures coverage of states reachable by **any** deviation, including irrational ones that would be exploitable by a Nash equilibrium opponent.

Hierarchy

$$\mathcal{C}(\mu^E, \nu^E) \leq \mathcal{C}_{\max}.$$

$\mathcal{C}(\mu^E, \nu^E)$ can be finite while $\mathcal{C}_{\max}$ is infinite — a crucial difference.

15.6 Hardness of Non-Interactive MAIL

A natural hope was that non-interactive MAIL only needs $\mathcal{C}(\mu^E, \nu^E)$ to be finite, since this captures rational deviations. If true, Behavior Cloning (BC), directly mimicking the expert at every observed state, would suffice whenever rational deviations are covered.

This hope is false. The following result answers the open question in the negative:

Theorem: Lower Bound for Non-Interactive MAIL

For any non-interactive MAIL algorithm Alg, there exists a Markov game $\mathcal{G}$ such that achieving Nash-Gap $\leq \mathcal{O}(\varepsilon)$ requires an expert dataset of size:

$$N = \Omega\left(\frac{\mathcal{C}_{\max}}{\varepsilon^2}\right).$$

Moreover, there exist games where $\mathcal{C}(\mu^E, \nu^E) < \infty$ but $\mathcal{C}_{\max} = \infty$, in which case **no non-interactive algorithm succeeds**, regardless of sample size.

15.7 The Hardness Construction: A 3-State Game

The lower bound is established via a strikingly simple Markov game with only **3 states**, illustrated below.

Hardness Construction

States: s_1 (start), s_2 , s_3 .

Transition (controlled by player 2):

- From s_1 : action $b_1 \rightarrow s_2$, action $b_2 \rightarrow s_3$ (regardless of player 1's action).

Rewards: $r(s_1) = r(s_2) = 0$. In s_3 , players play a normal-form game (e.g. Matching Pennies) with Nash value $v \in \{v_1, v_2\}$ differing by ε .

Nash Equilibrium: Under (μ^E, ν^E) , player 2 plays b_1 , so trajectories visit only s_1 and s_2 . State s_3 is *never visited* in equilibrium. Thus $\mathcal{C}(\mu^E, \nu^E) < \infty$.

The trap: A non-interactive learner never observes expert play in s_3 . Player 2 can deviate *irrationally* by playing b_2 (which would be exploitable by any Nash-playing opponent), reaching s_3 where the learner has no expert guidance. Since the learner cannot distinguish v_1 from v_2 without $\Omega(\varepsilon^{-2})$ samples from s_3 , and those samples never appear in the dataset, the learner fails.

Irrational vs. Rational Deviations

The key insight: $\mathcal{C}(\mu^E, \nu^E)$ only covers **rational** deviations (best responses to the expert). But in MAIL, the *learner's output* $(\hat{\mu}, \hat{\nu})$ may not be an expert Nash, so an adversary can deviate **irrationally** (in a way exploitable by a NE) to reach states the learner has never seen. Covering these irrational deviations requires $\mathcal{C}_{\max}$ to be bounded.

This is different from offline MARL where the reward is known: there, having the reward allows one to “fill in” uncovered states with pessimistic values, preventing exploitation. Without reward, no such defence exists.

16 Overall Summary

Big Picture

This lecture developed the theory of multi-agent reinforcement learning from first principles, moving from one-shot games through sequential games, and then from the online to the offline and imitation settings.

Game Theory Foundations

Normal form games capture one-shot simultaneous play via a payoff matrix. The key solution concept is the Nash Equilibrium (NE): a strategy profile from which no player benefits by unilateral deviation. In two-player zero-sum games, the von Neumann Minimax Theorem guarantees existence and uniqueness of the minimax value, and a mixed-strategy NE always exists. Correlated and coarse correlated equilibria (CE, CCE) generalise NE by allowing coordination; $\text{Nash} \subseteq \text{CE} \subseteq \text{CCE}$. Markov games extend MDPs to multiple agents: transitions and rewards depend on both players' actions, and the optimal Bellman equation replaces $\max_a$ with a matrix-game solve $\min_{\nu} \max_{\mu}$.

Learning Algorithms for Markov Games

Best-response (BR) based algorithms reduce MARL to a sequence of single-agent RL problems. Self-play cycles and does not converge. Fictitious play (best-response to the empirical average opponent) converges asymptotically in zero-sum games (Robinson 1951) but may require exponential samples. Double Oracle / PSRO are more efficient but treat policies as atomic actions,

yielding doubly-exponential guarantees when extended to Markov games. Dynamic programming approaches (Nash-VI; Optimistic Nash-VI) directly exploit Markov structure. Because computing a Nash of a general-sum game is PPAD-hard, Optimistic Nash-VI uses a CCE oracle instead — valid because in zero-sum games the marginals of a CCE are Nash.

Offline MARL

In the offline setting, the agent has only a fixed dataset and cannot re-query the environment. In single-agent offline RL, *single policy concentration* (covering only π^*) is necessary and sufficient, and pessimism (LCB-VI) achieves sample complexity $\tilde{O}(SC^*H^3/\varepsilon^2)$. This analogy fails for MARL: covering only the Nash trajectory (μ^*, ν^*) is *insufficient*, because verifying equilibrium requires evaluating all potential unilateral deviations. The necessary and sufficient condition is **unilateral concentration**: the dataset must cover every pair (μ, ν^*) and (μ^*, ν) . Under this assumption, Pessimistic Nash-VI (which applies opposite-direction bonuses to the two players and uses a CCE oracle) achieves polynomial sample complexity.

Multi-Agent Imitation Learning

Multi-agent imitation learning (MAIL) removes the reward signal entirely: the learner observes expert Nash trajectories but must recover an approximate NE without knowing payoffs. This is strictly harder than offline MARL. Even when the expert concentrability $\mathcal{C}(\mu^E, \nu^E)$ is finite (rational deviations are covered), a non-interactive learner can be forced to fail on states reachable by *irrational* deviations, states never visited in equilibrium. The correct coverage quantity is $\mathcal{C}_{\max}$, and the lower bound of $\Omega(\mathcal{C}_{\max}/\varepsilon^2)$ shows this is unavoidable. Behavior Cloning (BC) matches this lower bound up to a polynomial factor in $\mathcal{C}_{\max}$, making it rate-optimal among non-interactive algorithms.

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